HippoCortex

Biologically-Inspired Generative Replay in State Space Models for Exemplar-Free Continual Learning

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1. Research Problem

- Modern AI systems must learn from **continuous streams of data**
- Standard neural networks suffer from

  Catastrophic Forgetting
- Learning a new task destroys previously learned knowledge
- Existing solutions rely on **Experience Replay**
  - store past data
  - retrain repeatedly
- This leads to
  - high storage cost
  - privacy risks
  - poor scalability
- **Goal:** Continual learning without storing raw data

2. Transformers & Catastrophic Forgetting

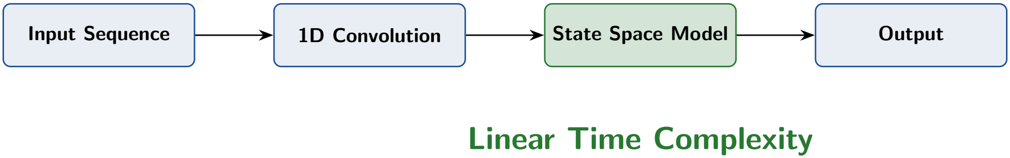
- Most foundation models rely on the **Transformer architecture**
- Transformers learn powerful representations using **self-attention**
- However they struggle with

  Catastrophic Forgetting
- Current workaround: **Experience Replay**
  - store thousands of past samples
  - retrain the model continuously
- Causes large memory usage and privacy concerns

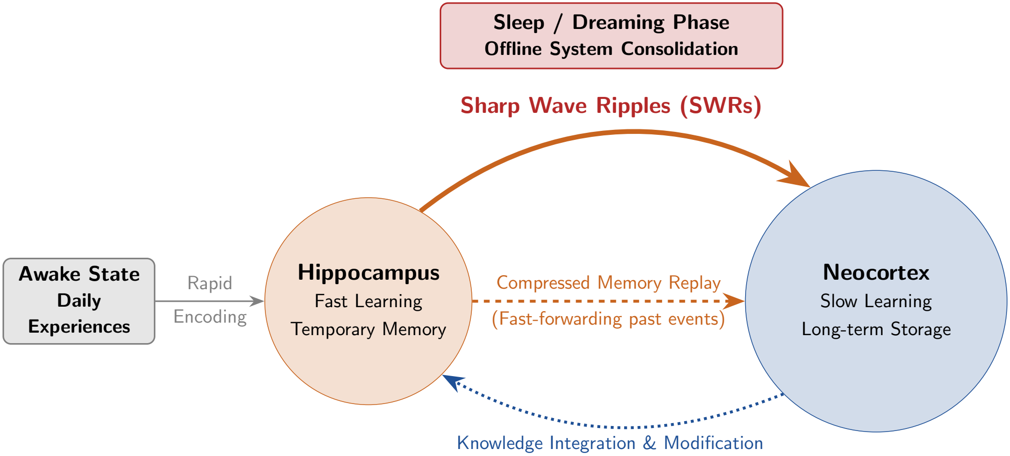
3. Transformer Bottlenecks

Quadratic scaling becomes expensive for long sequences

4. Shift to Mamba Architecture



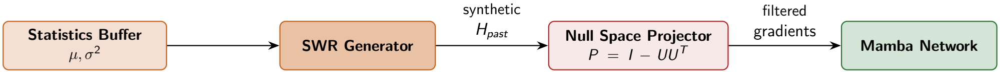
5. Biological Inspiration



6. Our Contributions

- Propose a new continual learning framework called **HippoCortex**
- Introduce **SWR Generative Replay**
- Generate synthetic hidden states instead of storing raw data
- Combine replay with **Null-Space Gradient Projection**
- Achieve continual learning with **constant memory footprint**
- Designed for **privacy-preserving edge AI systems**

7. HippoCortex Architecture



8. Method Comparison

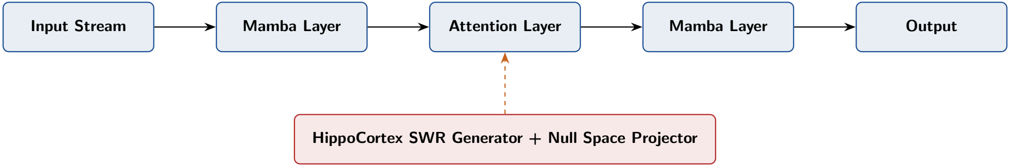
Method	Replay	Raw Data	Memory Cost
Transformer CL	Hard Replay	Yes	High
Mamba-CL (2024)	No	No	Constant
Inf-SSM (2025)	No	No	Constant
HippoCortex (Ours)	**SWR Replay**	**No**	**Constant**

Synthetic replay enables continual learning without storing past samples.

9. Experimental Setup

- **Datasets**
  - Split-CIFAR100
  - ImageNet-R
- **Baselines**
  - Naive Fine-Tuning
  - Mamba-CL
- **Evaluation Metrics**
  - Average Accuracy
  - Forgetting Rate
  - Memory Consumption
- Expected improvement: **∼8–10% accuracy gain over Mamba-CL (2024) and ∼4% over Inf-SSM (2025)**

10. Future Hybrid Architectures



11. Conclusion

Problem

- Continual learning causes catastrophic forgetting
- Current methods require storing past data

Solution

- HippoCortex architecture
- SWR generative replay
- Null-space gradient projection

Impact

- Privacy-preserving AI
- Constant memory footprint
- Edge AI lifelong learning